25900367

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCECC202 - NETWORKING FUNDAMENTALS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

ANSWER SCHEME

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries 2 marks

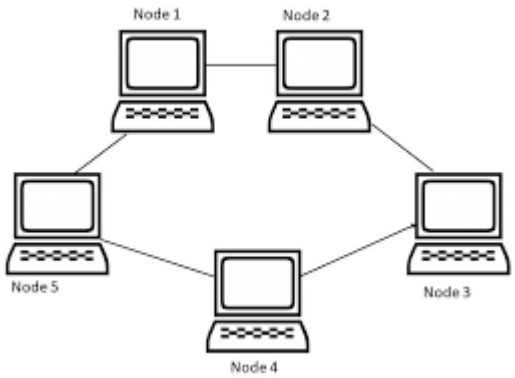
1.State the unit of bandwidth.

Explanation-2 mark

The unit of bandwidth is bits per second (bps) for digital signals, which can be scaled to higher units like megabits per second (Mbps) and gigabits per second (Gbps). For analog signals, bandwidth is measured in Hertz (Hz), which represents the range of frequencies a channel can transmit.

2. Show how five computers can be connected in a ring topology.

Diagram-2 mark



3. Describe bit interval and bit rate.

Explanation-2 mark

Bit interval is the time it takes to transmit a single bit, while bit rate is the number of bits transmitted per second. In other words, bit rate is the inverse of the bit interval, as it measures how many of those time slots for a single bit fit into one second. Bit rate is

typically expressed in bits per second (bps), with common prefixes like kilobits per second (Kbps) and megabits per second (Mbps).

4. Differentiate between attenuation and distortion.

Explanation-2 mark

Attenuation	Distortion
-Loss in signal strength because of resistance in the medium	- Any alteration of the original signal induced by any types of interference
-Does not change the waveform	- Changes waveform
-Relatively easy to overcome the effects	-Difficult to remove the effectes
-Amplitude reduces over the specific amount in the signal	-Attenuation happening at different places in the signal

5. What is meant by virtual-circuit network?

Explanation-2 mark

Virtual Circuit is the computer network providing connection-oriented service. It is a connection-oriented network. In virtual circuit resource are reserve for the time interval of data transmission between two nodes. This network is a highly reliable medium of transfer. Virtual circuits are costly to implement.

6. Differentiate between Stop-and-Wait ARQ and Go-Back-N ARQ with respect to any one key feature.

Any two points-1 mark for each

Stop and Wait	GoBackN
In the Stop and Wait protocol, the sender sends one frame and then waits for the receiver to acknowledge it.	Any frames that were sent after the frame that is believed to have been damaged or lost are resent.
In the Stop and Wait protocol, the Sender window size is 1.	In the Go-Back-N protocol, the Sender window size is N.
In the Stop and Wait protocol, the receiver window size is 1.	In the Go-Back-N protocol, the receiver window size is 1.
No particular sequence needs to be followed at the receiver end of the Stop and Wait protocol.	The Go-Back-N protocol's receiver end will only accept the correct sequence.
In the Stop and Wait protocol, the transmission type is Half duplex.	In the Go-Back-N protocol, the transmission type is full duplex.
Efficiency formula for the Stop and Wait protocol is 1/(1+2*a), where a is the ratio of propagation delay to transmission delay.	Efficiency formula for the Go-Back-N protocol is N/(1+2*a), where a is the proportion of propagation delay to transmission delay and N is the number of packets transmitted.

7. Recall which protocol is considered the basis for Ethernet.

Explanation-2 mark

Ethernet uses the IEEE 802.3 standard which defines protocols for the Physical and Data Link layers of the OSI model. On the Data Link layer, it splits into the Logical Link Control (LLC) and Media Access Control (MAC) layers, with the MAC layer using a mechanism called CSMA/CD (Carrier Sense Multiple Access with Collision Detection) to manage how devices share the network.

8. Differentiate between ACK and NACK with respect to error control.

Any two points-1 mark for each

Feature	ACK	NACK
Meaning	Data received successfully.	Data received with errors or not received.
Sender's Action	Sends next data packet.	Retransmits the problematic data packet.
Significance	Confirms successful transmission.	Requests retransmission for error recovery.
Impact on Flow	Promotes continuous data flow.	Interrupts flow to address errors.

9. State how many layers are in OSI.

Explanation-2 mark

There are 7 layers in the Open Systems Interconnection (OSI) model [1].

Physical Layer

Data Link Layer

Network Layer

Transport Layer

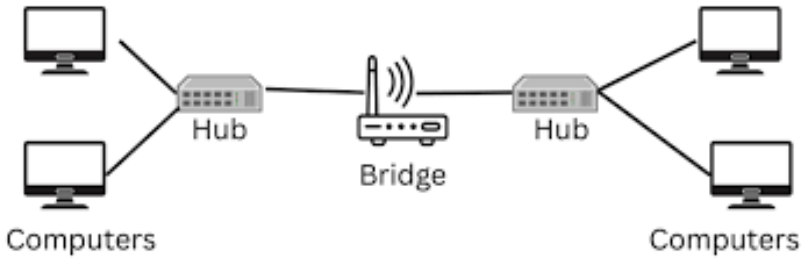
Session Layer

Presentation Layer

Application Layer

10. Draw a simple diagram showing how a bridge connects two LAN segments.

Diagram-2 mark



Part B

Short Answer Questions

Answer any 6 questions

Each question carries 5 marks

11. Compare FTP and HTTP for file transfer in simple terms.

Any five points-1 mark for each

PARAMETER	FTP	HTTP
Full Form	File Transfer Protocol	Hyper Text Transfer Protocol
Port Number	TCP port no 20 and 21	TCP port no 80 and 8080
RFC	RFC959, RFC765, RFC1738	RFC2616, RFC7230 and RFC7231
Philosophy	Used to transfer files from remote computer after connection is established	Used to transfer web pages from remote server after internet connection is established.
Communication	2-way communication system where upload and download of files from client to server can occur.	1-way communication system where content including pictures and texts can be transferred from server to client.
Speed of download	FTP is slower than HTTP	HTTP is faster than FTP when downloading one big file. HTTP can use parallel chunk download which makes it 6x times faster than FTP
Applicability	Used to access and transfer files.	HTTP is used to view websites.
Client	FTP can be accessed via the command line or a GUI	The common HTTP client is the browser
Usage	Used by fewer people	Most widely used
Authentication	Requires username and password for authentication	Does not requires username and password for authentication

12. Explain three types of guided media.

Explanation-5 mark

The three main types of guided media are **twisted-pair cable, coaxial cable, and fiber-optic cable**, all of which use physical cables to direct signals along a path. Twisted-pair and coaxial cables transmit signals as electrical currents, while fiber-optic cables transmit data as light signals.

13. Name and briefly describe the function of at least three common application layer protocols.

Explanation-5 mark

HTTP (Hypertext Transfer Protocol): This is the foundation of data communication for the World Wide Web. It is used by web browsers to request and receive web pages, images, and other resources from web servers, allowing users to browse the internet.

SMTP (Simple Mail Transfer Protocol): This protocol is essential for email services and is used to send and forward email messages from a client to a mail server, as well as between different mail servers across the internet.

DNS (Domain Name System): The DNS provides the "internet's phonebook" service, translating human-readable domain names (e.g., www.example.com) into numerical IP addresses that computers use to locate and connect with each other.

FTP (File Transfer Protocol): This protocol is used for transferring files between a client and a server over a network. It allows users to upload, download, and manage files on a remote system, using a separate control connection and data connection.

14. Indicate the IEEE standard used for Gigabit Ethernet and mention the type of cabling it supports.

Explanation-5 mark

IEEE 802.3ab: Defines Gigabit Ethernet over copper, often called 1000BASE-T.

IEEE 802.3z: Defines Gigabit Ethernet over fiber, known as 1000BASE-X.

15. Differentiate between stop - and- wait protocol and sliding window protocol.

Any five points-1 mark for each

Stop-and-Wait Protocol	Sliding Window Protocol
In Stop-and-Wait Protocol, sender sends one frame and wait for acknowledgment from receiver side.	In sliding window protocol, sender sends more than one frame to the receiver side and re-transmits the frame(s) which is/are damaged or suspected.
Efficiency of Stop-and-Wait Protocol is not that much good.	Efficiency of sliding window protocol is better than stop and wait protocol.
Sender window size of Stop-and-Wait Protocol is 1.	Sender window size of sliding window protocol is N.
Receiver window size of Stop-and-Wait Protocol is 1.	Receiver window size of sliding window protocol may be 1 or N.
In Stop-and-Wait Protocol, sorting is not necessary.	In sliding window protocol, sorting may be or may not be necessary.
Efficiency of Stop-and-Wait Protocol is 1/(1+2*a)	Efficiency of sliding window protocol is N/(1+2*a)
Stop-and-Wait Protocol is half duplex.	Sliding window protocol is full duplex.

Stop-and-Wait Protocol	Sliding Window Protocol
Stop-and-Wait Protocol is mostly used in low speed and error free network.	Sliding window protocol is mostly used in high speed and error-prone network.
In Stop-and-Wait Protocol, the sender cannot send any new frames until it receives an acknowledgment for the previous frame.	In sliding window protocol, the sender can continue to send new frames even if some of the earlier frames have not yet been acknowledged.
Stop-and-Wait Protocol has a lower throughput as it has a lot of idle time while waiting for the acknowledgment.	Sliding window protocol has a higher throughput as it allows for continuous transmission of frames.

16. Explain how a firewall can prevent unauthorized access.

Explanation-5 mark

A firewall prevents unauthorized access by acting as a barrier between networks, inspecting incoming and outgoing data packets, and blocking any that don't meet predefined security rules. It examines traffic based on criteria like IP addresses, ports, and protocols, allowing only legitimate traffic to pass while blocking malicious or suspicious activity to protect the network from threats like malware, hackers, and other cyberattacks.

17. Explain any two techniques used for congestion control.

Explanation-2.5 mark for each

Two techniques for congestion control are open-loop and closed-loop methods. Open-loop techniques prevent congestion before it happens, often through rules like slow start or leaky bucket algorithms to manage transmission rates. Closed-loop techniques, on the other hand, are reactive and take action after congestion is detected, using feedback mechanisms like choke packets or explicit congestion notification to signal a source to slow down.

18. Distinguish between static IP addressing and dynamic IP addressing in both IPv4 and IPv6.

Any five points-1 mark for each

IPv4	Ipv6
Addresses are 32 bits (4 bytes) in length.	Addresses are 128 bits (16 bytes) in length
Deployed in 1981	Deployed in 1999
Address format is dotted Decimal Notation	Address format is Hexadecimal Notation
Both routers and the sending host fragment packets.	Routers do not support packet fragmentation. Sending host fragments packets
Header includes a checksum.	Header does not include a checksum.
Header includes options.	Optional data is supported as extension headers.
Broadcast addresses are used to send traffic to all nodes on a subnet.	IPv6 uses a link-local scope all-nodes multicast address.
Configured either manually or through DHCP.	Does not require manual configuration or DHCP.

Part C

Essay Questions

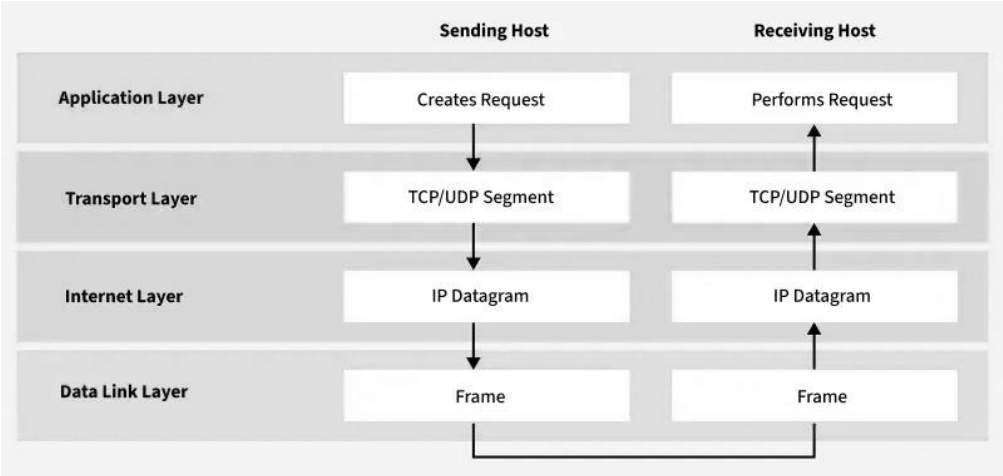
Answer any 2 questions

Each question carries 10 marks

19. Analyse the TCP/IP model by explaining the functions of each layer and associating common protocols with them.

Diagram-5 mark

Explanation-5 mark



20. Given a wireless system with limited bandwidth, how would you apply spread spectrum techniques to ensure that multiple users can access the network efficiently?

Explanation with diagram-10 mark

To enable efficient multi-user access with spread spectrum in a limited bandwidth, Code Division Multiple Access (CDMA) is used by assigning a unique spreading code to each user, allowing them to share the same bandwidth simultaneously. The two main spread spectrum techniques are Frequency Hopping (FHSS), which rapidly changes the frequency of the signal, and Direct Sequence (DSSS), which modulates the signal with a pseudo-random code.

21. Demonstrate with the help of a block diagram how a call is established and maintained in a cellular network, including the role of base stations and mobile switching centers.

Diagram-5 mark

Explanation-5 mark

